

Metals and non-metals

Student: Class:

Classifying elements

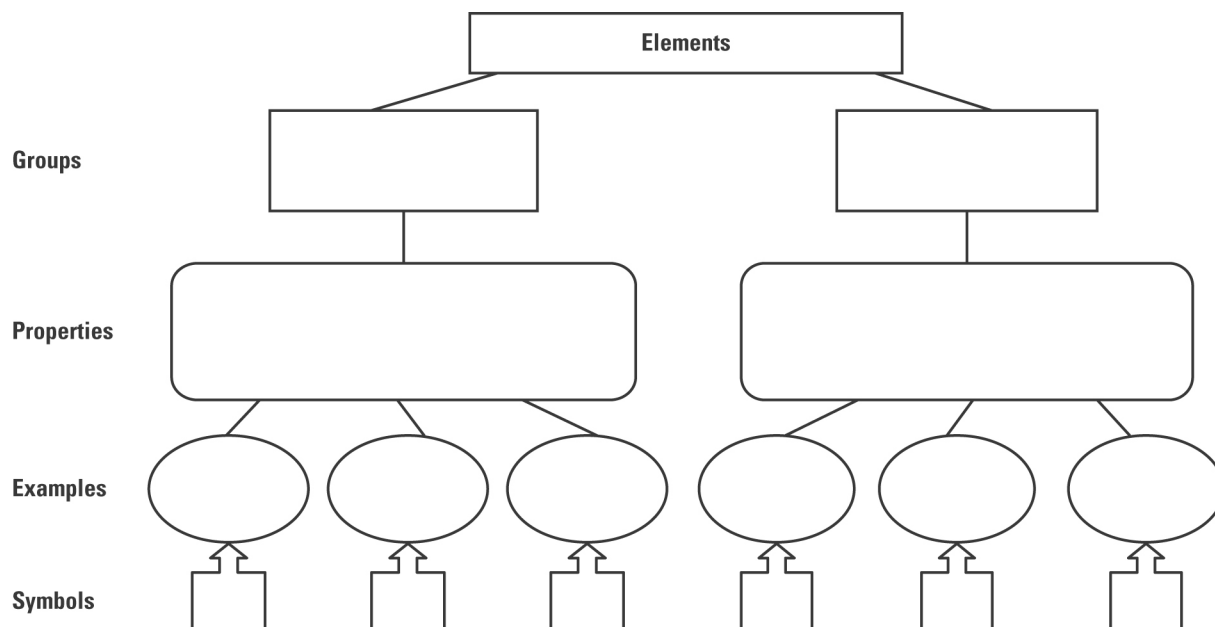
1. Complete the following diagram by inserting words and phrases from each section of the lists below:

Groups: metals, non-metals

Properties: dull, lustrous, good heat conductor, poor heat conductor, poor electricity conductor, good electricity conductor, brittle, malleable, high melting point, low melting point

Examples: iodine, zinc, fluorine, copper, phosphorus, silver

Symbols: F Ag Zn P I Cu



Metalloids

2. Not all elements readily fit in the metal or non-metal groupings. These elements are called metalloids. There is some disagreement about the number of metalloids. The metalloids are: boron, silicon, germanium, arsenic, antimony, tellurium and astatine (polonium is classified as a metal rather than a metalloid by many authorities). Examine the table below and classify W, X and Y as metals, metalloids or non-metals.

W: X: Y:

Property	W	X	Y
Lustre	Moderately high	Low	High
Electrical conductivity	Low	Nil	High
Density	Medium	Low	High
Melting point	Very high	Low	High